

The Mega2R R merge package: tools for accessing and processing common genetic data formats in R

Robert V. Baron¹, Justin R. Stickel¹, and Daniel E. Weeks^{1,2,*}

¹Department of Human Genetics, Graduate School of Public Health, University of Pittsburgh, Pittsburgh, Pennsylvania, 15261, USA

²Department of Biostatistics, Graduate School of Public Health, University of Pittsburgh, Pittsburgh, Pennsylvania, 15261, USA

*Corresponding author

Email: Robert V. Baron - rvb5@pitt.edu, Justin R. Stickel - jrs110@pitt.edu, Daniel E. Weeks - weeks@pitt.edu

Abstract

The standalone C++ Mega2 program has been facilitating data-reformatting for linkage and association analysis programs since 2000. Support for more analysis programs have been added over time. Currently, Mega2 converts data from several different genetic analysis data formats (including PLINK, VCF, BCF, IMPUTE2) into the specific data requirements for over 40 commonly used linkage and association analysis programs (including Mendel, Merlin, Morgan, SHAPEIT, ROADTRIPS, MaCH/minimac3). Recently, Mega2 has been enhanced to use a SQLite database as an intermediate data representation. Additionally, Mega2 now stores biallelic genotype data in a highly compressed form, much like that of the GenABEL R facility and the PLINK binary format.

Our Mega2R package now makes it easy to load SQLite3 Mega2 databases directly into R as data frames. In addition, we have developed C++ functions for R to decompress needed subsets of the genotype data in a memory efficient manner. In addition there are extended functions that illustrate how to use the data frames as well as perform useful functions: the Mega2pedgene function to run the pedgene R package to carry out gene-based association tests on family data using selected marker subsets, the Mega2VCF function to output the Mega2R data as a VCF file and related files (for phenotype and family data), and the Mega2GenABEL function to convert the data frames into GenABEL R `gwa.data-class` objects. The Mega2 program and the Mega2R R package are both open source and are freely available, along with extensive documentation and a 'quick-start' tutorial, from our <https://watson.hgen.pitt.edu/register> web site.

Keywords

database, genotypes, genome-wide association studies, linkage, Mega2, phenotypes, R, SQLite3

Introduction

During an association or linkage analysis project, one may need to analyze the data with several different programs. Unfortunately, it can often be quite difficult to get one's data in the proper format desired by each different computer program. Not only must the data be converted to the proper format, but also the loci must be reordered into the desired order. Writing custom reformatting scripts can be error-prone and very time-consuming. To address these problems, we created Mega2 [1, 2], which can be obtained from <https://watson.hgen.pitt.edu/register/> or via BitBucket <https://bitbucket.org/dweeks/mega2>. The early Mega2 could read input data in only a few formats: LINKAGE format [3, 4, 5] and Mega2 format. The Mega2 format allowed you to specify additional information in a more straightforward way, via a simple tabular format, than can be done with LINKAGE format. The earliest versions of Mega2 were written in the C computer language.

Over time, Mega2 was upgraded to the C++ computer language and some of the internals were rewritten. Reformatting data for more analysis programs was gradually added to Mega2. The data formats for genetic data changed over time; and Mega2 was enhanced to read input data in PLINK format [6], VCF format [7], and IMPUTE2 format [8, 9, 10, 11]. The volume of genome-wide marker data increased significantly, so Mega2 was extended to support the compression of biallelic genotype data (though still supporting microsatellite and other polymorphic data as non-compressed legacy data). As the volume of genotype data increased, it became slow and tedious to validate the genotype data and generate allele frequency data for each separate analysis that Mega2 was used for. Rather, Mega2 was restructured so that the validation and meta data generation were performed once for a given data set.

Initially, Mega2 used C (language) structures to store its intermediate data before analysis. Several alternative strategies were considered to dump these data for subsequent reuse without having to recompute, reanalyze, and revalidate the data. The first choice was to dump the raw structure data to disk. Reloading the data is fast but it is much harder to inspect the data for debugging. Also, the internal pointers would have to be relocated to new storage areas. A better solution, for many reasons, was to create a SQLite (<https://www.sqlite.org/>) table for each C structure and “insert” the C structure data into the table. Mega2 uses this database, via a menu-driven interface, to provide the data needed for any particular analysis. The same database provides the data for all the different analyses.

In addition, the data are inherently inspectable and there are database tools to help to selectively extract data. The data are portable and can be shared on different platforms. Since interfaces from many languages are provided for SQLite, the data are accessible in many languages besides C. Of particular interest is R [12] because there is a wealth of existing genetic analysis programs already created, well documented, and available in CRAN and Bioconductor. Also, researchers regularly use R to develop new analysis algorithms because R has been used in the past to develop statistical programs and has many useful libraries. We describe here our Mega2R package, which makes it easy to load SQLite3 Mega2 databases directly into R as data frames for further analysis and manipulation. In addition we document several R functions that illustrate how to use the Mega2R data frames as well as perform useful functions: the Mega2pedgene function to run the pedgene R package to carry out gene-based association tests on family data using

selected marker subsets, the `Mega2VCF` function to output the Mega2R data as a VCF file and related files (for phenotype and family data), and the `Mega2GenABEL` function to convert the data frames into GenABEL R `gwaa.data-class` objects [13].

Methods

Environments

Before we discuss the Mega2R package, we must describe one important convention of how the database information is stored and how it is accessed later. The Mega2R R package function, `read.Mega2DB`, reads the tables needed by Mega2R from a Mega2 SQLite database into a collection of data frames. It returns an “R environment” containing these data frames. (If you are unfamiliar with environments, you can think of them as data frames. `'ENV'${locus_table}` will access the `locus_table` variable from the environment, `'ENV'`, similar to fetching an “observation” from a data frame. The difference is when you change data in a data frame passed to a function, the change does not affect the original data frame. Only the function’s local value is changed; ALL changes are forgotten when the function exits. If you change the data in an environment passed to a function, the change is permanent.) All Mega2R functions that do not return an environment need to have an environment supplied as an argument; the environment is used to store the data frames that contain the SQLite database. There are two ways to pass the environment. If you assigned the result of `read.Mega2DB` to the variable `'seqsimr'`, then you could supply the value `'seqsimr'` as the named argument, `'envir'`:

```
showMega2ENV(envir = seqsimr)
```

The second choice is a bit of a “hack” but it is very convenient. Every Mega2R function (that does not return an environment) has a named argument, `'envir'`, defined to take on the default value `'ENV'`:

```
envir = ENV
```

as in

```
showMega2ENV = function(envir = ENV) { ... }
```

The code above, assigns the local variable, `'envir'` the default value `'ENV'`. Thus if `'envir'` is not provided in the function call, R will use the default assignment and look up the value of `'ENV'`. Hopefully, `'ENV'` was set as the return value of a call to `read.Mega2DB`.

Mega2 SQLite Database

In the latest Mega2 design, there are SQL tables for loci data, marker data, allele data, map data, phenotype data and genotype data as well as pedigree and person data. For completeness, Appendix A shows all the tables and their fields. Here, we will make a few general observations about the tables. The first issue was mentioned earlier: What should we do with pointers in C++. Where necessary, a field was added to a table,

we call a “link” field, viz. `locus_link`. This field is a unique monotonic integer. When the data are read back in via C++, these links are looked up to find a pointer to use. Consider the **locus_table** and the **allele_table**. They both have a `locus_link`. The link in the **locus_table** is hashed to yield a pointer to the corresponding entry in the **locus_table**. When the **allele_table** is read, the `locus_link` is looked up, then a pointer is inserted into a new field in the “in memory” **locus_table** (using the look up pointer) to point to the memory location of the **allele_table** entry, i.e. the **locus_table** in memory will point to the **allele_table**. These “links” are also very useful in R for performing merges of related tables. For example, in the above case, merge can produce a table with rows containing both locus and allele data. Similar links are used to associate family data with person data, as well as associate person data with phenotype and genotype data.

Most tables are composed of integer, numeric and string data. The phenotype table is an exception. The normal trick of adding an extra column per row (the phenotype id) to indicate which phenotype’s data is contained becomes more complicated because affection and quantitative data have different data types and number of values. Instead, the phenotype data for each person is one long byte vector composed of multiple values with 8 bytes of data per value either representing a quantitative value or an affection status value. Similarly, the genotype data are a raw vector with a pair of bytes representing a genotype for non biallelic markers and a raw compressed bit vector with two bits per genotype for biallelic markers. Note: the original compressed genotype vector created by Mega2 is one long vector. The genotype data written to the database is split into separate vectors for each chromosome. The end of each vector may have 0, 1, 2, or 3 unused bit pairs representing no markers. The Mega2 C++ program reassembles the vectors without these gaps. In addition, it can select specific chromosomes and/or specific markers needed for analysis and so does not need to store all the genotype bit vector data into active memory. The R version is not as sophisticated; it reassembles all the chromosome vectors and includes these gaps.

Mega2R Package Base Functions

The function, `read.Mega2DB` loads the Mega2 SQLite database into R data frames.

```
library(Mega2R)
ENV = read.Mega2DB("seqsimr.db", verbose = TRUE)
```

If the verbose flag is true, summary statistics will be printed as the data frames are created.

Data frames ending in “_table”, viz. **person_table**, **locus_table**, contain raw data copied directly from the SQLite database, although some of the original fields that are only needed for C++ Mega2 are not copied.

Other data frames in the R environment contain derived information. The function, `mkfam`, is called by from `read.Mega2DB` to create the **fam** data frame, the equivalent of a PLINK .fam file. It merges the rows from the **pedigree_table**, **person_table**, and **phenotype_table** to make a **fam** data frame. The **pedigree_table** has a `pedigree_link`, **person_table** has a `pedigree_link` and `person_link`, and **phenotype_table** has a `person_link` field. These link fields are used by the R ‘merge’ function to assemble the data. (Note: The **genotype_table** also has a `person_link` field. Note: The analog to these person “_link” fields for marker data frames is the `locus_link` field.) Later, you may want to prune the **fam** data frame to restrict the samples processed. For,

example you might want to eliminate samples with unavailable case/control status. After you prune the **fam** information, the function, *setfam*, will set the **fam** data frame. It will also prune the **phenotype_table** and **genotype_table** to contain only those persons left in the **fam** data frame.

The **unified_genotype_table** collects all the raw byte vectors from the **genotype_table** by chromosome for each person and concatenates them into a new vector for each person. The concatenation can introduce up to three “virtual markers” at the end of each chromosome if the number of markers contained is not an exact multiple of 4 (genotypes per byte). (Note: Creating the **unified_genotype_table** with no “virtual markers” would require a lot of bit manipulations in R which would not be terribly efficient; while keeping track of the virtual markers introduced by the gaps is not particularly hard. The Mega2 program written C++ does bit manipulations to just copy the bits for the markers it will need for later analysis.)

The **marker_table** contains the offset of the marker genotype data in the **genotype_table** data vector column assuming all the markers across all the chromosomes are contiguous. There is an analogous offset column needed for the **unified_genotype_table** accounting for the “virtual markers” in the gaps; this offset column is computed and added to the **marker_table**. The **marker** data frame is a subset of the **marker_table** data containing only the name, chromosome, position, original **marker_table** offset and marker offset into the **unified_genotype_table** vector.

Mega2R Marker Range Functions

We might have genotype data for thousands of samples and millions of markers in a study. But we are unlikely to have the memory to support this large a matrix in R. The Mega2 genotype data is compressed in the SQLite database and remains compressed in its data frame. Most of the time, we do not need to decompress all of it at once. We typically want to examine the markers of a chromosome or of some genes of interest. We only decompress the set of markers we need. The key idea is we want to find the set of markers that fall in some base pair range and process just those markers. We will introduce two ways to specify these ranges. In addition, we want to iterate through multiple ranges as would appear in one or more gene transcripts and process the markers. Finally, we process a each range of markers by invoking a callback function on the range, markers and related data.

setRanges and *applyFnToRanges* function

Mega2R has an internal list of the chromosome and base pair ranges for many gene transcripts. These transcripts come from the UCSC Genome Browser reference assembly GRCH37. The list was further modified to eliminate multiple records of the same gene with the exact same transcript start and transcript end. The list contains about 29,000 records. But there are several reasons that default transcript ranges may not be to your liking. Sometimes the transcripts for a gene appear to have overlapping start/end position. The transcripts have one start and one end but may not account for non coding regions, promoters, etc. Perhaps you would prefer to specify a transcript names and for each name provide a start position, end position and chromosome. The *setRanges* function lets you store custom ranges; it has two arguments: a range data frame that has at least the start position, end position and chromosome “observations” and a

selector vector of 3 integer that indicates which observation variable in the range data frame contains the chromosome, start position and end position.

The *applyFnToRanges* function (and later *applyFnToGenes* function) take an initial argument that is a callback function that is called repeatedly for each transcript that has markers present. They also take a final argument that is an “R environment”. *applyFnToRanges*, with no additional arguments, will iterate through each row of the Mega2R ranges, determine the markers contained between the start and end, compute the genotype matrix, and then invoke a callback function. Alternatively, *applyFnToRanges* may be given a range argument and a selector argument which will cause it to use the ranges supplied rather than the default. These two arguments are the same as provided to *setRanges*.

applyFnToRanges requires a call back function of four arguments: the first is a matrix of genotypes for all the markers in the transcript, the second is a data frame of the markers in the transcript, the third is a single row from the data frame that defines the transcript (minimally it has a name, ID, start position, end position, and chromosome), and the final argument is the “R environment” that contains all the Mega2R data frames. The call back function can apply whatever analysis is required and it can access the other data frames available in the “R environment” and even store intermediate results back into the “R environment”.

setAnnotations and applyFnToGenes function

Mega2R also loads a list of genes and their transcripts from the Bioconductor data Annotations: "org.Hs.eg.db" and "TxDb.Hsapiens.UCSC.hg19.knownGene". The former translates a gene name or alias to the entrez ID of the gene. The latter gives start position, end position and chromosome for each transcript known for the each gene. You might want to use the transcript data from another build that is available at Bioconductor or follow their instructions make your own mapping. The Mega2R function, *setAnnotations*, lets you supply alternate string names for these two tables. Mega2R will load the selected tables when you access the *applyFnToGenes* function.

applyFnToGenes specifies a list of genes from which to extract the transcripts and the corresponding ranges. In addition, *applyFnToGenes* can take a number named arguments that specify other ways to indicate transcript ranges: on sub-ranges of chromosomes (*ranges_arg* argument), on the entire chromosome (*chrs_arg* argument), and/or on an arbitrary list of markers (*markers_args* argument). The call back function of four arguments and list of ranges is passed to *applyFnToRanges*. Note: If the special gene name, “*”, is passed as the only gene argument, all of the ranges for all of transcripts will be chosen.

applyFnToMarkers function

applyFnToMarkers is the simplest function that uses the aforementioned callback function. It takes one argument that is rows of the **marker** data frame. It creates a genotype matrix for these markers and then invokes the call back function with the genotype matrix, the marker argument data frame, NULL, and the “R environment”. (NULL because there is no range information.)

getgenotypes and getgenotypesraw functions

The functions, *getgenotypes* and *getgenotypesraw* return a matrix of nucleotide pairs or a matrix of encoded integers with a column for each marker and containing a row per sample. Both functions take a single argument which is rows of the **marker** data frame. This argument is computed by either the *applyFnToRanges*, *applyFnToGenes* or *applyFnToMarkers* function. The **marker** data frame has two offsets that are used internally by the decompression code; the **marker** also has a name, chromosome number and position that can be used to identify a marker to the user. The two *getgenotypes* functions are dispatches that call Rcpp code. They collect genotype and allele data from the “R environment” data frames and pass these data arguments to the Rcpp code; they also call the proper function for the level of compression: 2 bits or 2 bytes per marker. For each marker specified, *getgenotypes* will return for each sample the corresponding pair of nucleotides as a string, while *getgenotypesraw* will return the first allele integer value and the second allele integer value; the two integers are returned as a single integer with the integer for the first allele shifted 16 bits and or’ed to the integer for the second allele.

Mega2pedgene Function

The ‘pedgene’ package performs Gene-level association tests with disease status for pedigree data: kernel and burden association tests on family data, and is available on CRAN as (<https://CRAN.R-project.org/package=pedgene>). It was written by Daniel Schaid and Jason Sinnwell[14].

Mega2pedgene invokes *applyFnToRanges* on the default ranges or calls *applyFnToGenes* on gene names.

We call *init_pedgene* rather than *read.Mega2DB*; the former uses the function in the Mega2R package to just load a Mega2 database (*dbmega2_import*). Then it generates a modified the **fam** data frame; it must eliminate samples where the case/control is not known which is required by ‘pedgene’. In addition, *init_pedgene* computes and saves some state into the environment to be used later. The CRAN package, *pedgene*, requires an encoding of the genotypes 1/1, 1/2 and 2/2 to 0, 1, and 2 respectively; plus the alleles must be flipped so that 1 is the allele with the major allele frequency. DOpedgene transforms its genotype input matrix and then it calls the *pedgene* function with several different marker weights. It returns the p-value of the kernel and burden association statistics for each weight. The results are stored in a data frame. The data frame will be written to a file if one were specified as the *filename* argument of *init_pedgene*.

Note: The call back function, *DOpedgene*, duplicates the code that we used in an earlier “manual run”. We had the results of running ‘pedgene’ on many transcript ranges for one of our private dataset and we got the same results running *Mega2pedgene* on the same dataset. But the original code harness was hand coded and hard to change.

Mega2SKAT Function

The SKAT[15, 16, 17] package also performs “kernel-regression-based association tests including Burden test, SKAT and SKAT-O. These methods aggregate individual SNP score statistics in a SNP set and efficiently compute SNP-set level p-values.”

The *Mega2SKAT* function is rather similar to *Mega2pedgene* (above). *init_SKAT* behaves very much like *init_pedgene*, in addition it computes a phenotype data frame. The *Mega2SKAT* function has the signature:

```
Mega2SKAT = function (f, ty, gs = 1:100, skat = SKAT::SKAT, envir = ENV, ...)
```

f and **ty** are the formula and phenotype type that are used to invoke *SKAT_Null_model*. **skat** indicates the particular SKAT function to call and the ... arguments are a place holder for any additional arguments that the particular **skat** needs.

Mega2VCF Function

The VCF[18] data format

(<http://www.internationalgenome.org/wiki/Analysis/Variant%20Call%20Format/vcf-variant-call-format-version-40/>) was originally defined by the 1000 Genomes Project[19]

(<http://www.internationalgenome.org/home>) for data storage. The current version of data format can be found at (<http://samtools.github.io/hts-specs/>)[20]. The *Mega2VCF* function serves several purposes. It allows VCF format files to be generated for analysis by R programs that require VCF input. It shows how to extract data provided in the *Mega2R* data frames for subsequent analysis by other R packages. Finally, it provides a regression for *Mega2R* by producing the same files from an SQLite database via R code that *Mega2* can produce from the identical database via C++ code. The VCF file and related files calculated both ways should be the same except for time stamps.

You must first call the *read.Mega2DB* function from the *Mega2R* package to load a *Mega2* database. A .vcf file is the main output of *Mega2VCF*; it is created from the genotype matrix, supplemented by columns from the **fam** table and **allele_table** table. Besides the .vcf file, *Mega2VCF* generates several additional files.

Below, we indicate the file suffix, the internal function that generates the file, and the file contents:

file suffix	function	contents
.vcf	Mega2VCF mkVCFhdr	contains the multi column VCF file with header; each row contains all the samples for that marker
.fam	mkVCFfam	contains the first 6 columns of the pedigree or .fam file
.freq	mkVCFfreq	contains the allele frequency information
.map	mkVCFmap	contains the supplied maps with their genetic or physical distances
.phe	mkphenotype	contains the phenotype information: quantitative and affection status (Note: This function was moved to the mega2rcreate.R file.)
.pen	mkVCFpen	contains the penetrance information

The code in each function illustrates how to assemble the corresponding data from the Mega2R data frames.

The *mkphenotype* is a bit subtle. The database contains a raw vector of 8 bytes * the number of phenotypes (per person). The 8 bytes are either one double precision float number or two integers depending on the encoding in the **locus_table** (quantitative or affection status). *readBin* is used to convert the raw bytes in each 8 byte vector to the correct form.

Another interesting aspect of the code design is how we compute genotype data for a large numbers of markers without needing large amount of memory. We build the *.vcf* file a chunk at a time, where a chunk contains a relatively small number of markers (currently 1000). We generate the genotype matrix for each chunk then append the results for the chunk (of markers) to the *.vcf* file. Then, we get a new chunk and repeat. This looping strategy might prove useful for other situations where the results can be written incrementally.

Mega2GenABEL Function

The 'GenABEL'[13] package is available on CRAN at (<https://CRAN.R-project.org/package=GenABEL>). The GenABEL CRAN package provides many functions for GWAS analysis and it accepts GWAS data in several formats. But Mega2 accepts input in still more formats, notably, PLINK bed, VCF, IMPUTE2 and even Linkage. Thus *Mega2GenABEL* can be a bridge to easily convert data for GenABEL analysis.

You must call the *read.Mega2DB* function from the Mega2R package to load your database and should save the returned 'environment' into 'ENV'. The function, *Mega2GenABEL*, returns a *gwaa.data-class* object for the selected markers. It operates by re-writing the Mega2R data frames into one of the input formats that GenABEL supports; currently, this is PLINK *.tped* format. This requires a *.tped* file, a *.fam* file, and a *.phe* file; they are stored in a temporary directory. The *.tped* file looks very much like a VCF *.vcf* file with entries in each row for the samples' genotype data for each marker. (But, in comparison to the *.vcf* file, there are fewer fields of genotype metadata in a *.tped* file and the genotype data specify real nucleotide labels rather than VCF's REF/ALT field references.) *Mega2GenABEL* calls the GenABEL "glue" functions, *convert.snp.tped*, to process the a *.tped* file and *.fam* file into a generic GenABEL "raw" format file, then it converts that raw format file and a phenotype file into a GenABEL *gwaa.data-class* object via the function, *load.gwaa.data*, and then it deletes the intermediate files.

The function *Mega2ENVGenABEL* is experimental but should produce same object as *Mega2GenABEL*. It does not write out files, but rather converts the Mega2R genotype and related data in memory to a *gwaa.data-class* object. It is mainly written in Rcpp and runs 10 to 20 times faster than the *Mega2GenABEL* function. A bit more testing and we will suggest *Mega2ENVGenABEL* be used exclusively.

The rest of this document shows some complete examples of Mega2R package use. First, we illustrate the base functions of the package, Mega2R. Then we demonstrate several extended functions, *Mega2pedgene*, *Mega2SKAT*, *Mega2VCF* and *Mega2GenABEL*. Note: Not all the arguments and details of the functions are discussed in these examples, if you need programming details, look at the Mega2R CRAN package documentation or the actual code in the git repository.

Use Case

The R code in the examples below is actually run during the creation of this document using the `knitr`[21, 22, 23] R package.

It is always difficult to share private study data. We have chosen to use the SeqSIMLA2 [24] program to generate a dataset for us. We will construct a small PLINK ped dataset using the simulator, which is available from https://sourceforge.net/projects/seqsimla/?source=typ_redirect. We started with the first example under the “prevalance” tab on the “Simulate by Disease status” tab on the tutorial page on the <http://seqsimla.nhri.org.tw/prevalence>. This simulation creates 1,380 samples of 500,000 markers; a bit too much for sharing. But a bit less than 200,000 markers are actually polymorphic while the rest are monomorphic; smaller but the files are still too big. We chose to subsample the latter down to 1,380 people and 1,000 markers. These data will be used to illustrate Mega2 and Mega2R operations that follow. Note: The simulator produces markers for only a subrange of chromosome 1.

Installation

R

We will assume that you have started an session at which you type the commands in the tutorial.

To run any of these exercises, you should install the package Mega2R.

```
install.packages("Mega2R")
```

Bioconductor

In Section below, we will carry out gene-based association tests, where ‘genes’ are defined according to a database containing the boundaries of the gene transcripts. This requires two Bioconductor Annotations databases to be installed. The first line (below) loads the Bioconductor loader and the next two lines install two annotation databases. One annotation database provides the gene transcript locations and the other maps gene names to entrez gene IDs. (Note: As described in Section , you may choose a different transcript database from Bioconductor or construct one of your own. If so, you should install it with *biocLite* as illustrated below.) Please type in R:

```
source("https://bioconductor.org/biocLite.R")
biocLite("TxDb.Hsapiens.UCSC.hg19.knownGene")
biocLite("org.Hs.eg.db")
```

The above steps are run once.

Tutorial Data

First, you will need to load the Mega2R package. Type:

```
library(Mega2R)
```

```
## NULL
```

The files you will need for this tutorial are provided in this package. You need to extract these files to the current directory. This is done by running:

```
dump_mega2rtutorial_data(".", " ")
```

You should see the following files:

MEGA2.BATCH.seqsimr	MEGA2.BATCH.srdta	MEGA2.BATCH.vcf
Mega2r.map	Mega2r.ped	mega2r.Rmd
mega2r.pdf	seqsimr.db	srdta.db

Note: The use of the "mega2" executable in our examples expects the Mega2.BATCH.<name> files to be in the working directory and the latter files expect their data files to be in the working directory. When you are done with these exercise, the "clean" command will remove these files and other temporary files:

```
clean_mega2rtutorial_data()
```

Using Mega2R to access and process genetic data

Creating a Mega2 database

We have provided files in this package that contain the data from the simulation. These files are in PLINK ped format data:

- Mega2r.ped
- Mega2r.map

If you do not wish to install Mega2 right now, you can use the seqsimr.db database that is in the tutorial directory.

You can obtain the Mega2 program from <https://watson.hgen.pitt.edu/register/>. Then, you will invoke Mega2 on your data. To make matters simple, we will use a pre-constructed Mega2 batch file to automate the processing by Mega2. To run Mega2 to process and create the 'SQLite' database 'seqsimr.db', we issue this command at the Unix prompt:

```
mega2 MEGA2.BATCH.seqsimr
```

Note: This document will not invoke *mega2*, but use the seqsimr.db database that is in this package.

Note: The vignette associated with the Mega2R CRAN package illustrates what this Mega2 run would look like.

If you do not provide the command-line argument giving the name of a BATCH file, Mega2 will proceed to ask a series of questions to collect the information needed to produce a database. In addition, it will create a Mega2.BATCH file, similar to the one we suggested you use. You can look at the "Quick Start" section of the Mega2 documentation https://watson.hgen.pitt.edu/docs/mega2_html/mega2.html to better understand the interactive process.

The MEGA2.BATCH.seqsimr file begins with a rather long comment indicating the keyword values that may be set and their default value. Toward the end of the file, we see the inputs set to **Mega2r.ped** and **Mega2r.map**, indicate the input is PLINK ped format with parameters, and indicate that Mega2 should produce a database called **seqsimr.db**, etc. (The initial comment section is not shown and unimportant lines are elided.) Notice that there is no analysis, just a database creation.)

```
Input_Database_Mode=1
Input_Format_Type=4
Input_Pedigree_File=Mega2r.ped
Input_PLINK_Map_File=Mega2r.map
...
PLINK_Args= --cM --missing-phenotype -9 --trait default
...
Value_Marker_Compression=1
Analysis_Option=Dump
...
DBfile_name=seqsimr.db
...
```

If you wish to use any of the Mega2R functions described here on your own data, you will have to run “mega2” to convert your data into an ‘SQLite’ database.

Reading and Examining a Mega2 Database

The Mega2R package facilitates reading genetic data from a Mega2-created ‘SQLite’ database.

Reading a Mega2 database After you have created the SQLite database, start up the R program. Load the Mega2R package, then use the function `read.Mega2DB` to read a Mega2 database.

```
library(Mega2R)
ENV = read.Mega2DB("seqsimr.db", verbose = TRUE)
```

The first argument should be the path to the database. Providing the optional argument, ‘verbose’, causes the read function to summarize the data frames created, their fields and their sizes. Finally, an ‘environment’, that contains the data frames is returned.

```

ENV = read.Mega2DB("seqsimr.db", verbose = TRUE)

## int_table 34 3
## int_table pId key value
##
## pedigree_table 20 8
## pedigree_table pId Num EntryCnt Name PedPre OriginalID origped pedigree_link
##
## person_table 1380 11
## person_table pId UniqueID OrigID FamName PerPre ID Father Mother Sex pedigree_link
person_link
##
## pedigree_brkloop_table 20 8
## pedigree_brkloop_table pId Num EntryCnt Name PedPre OriginalID origped pedigree_link
##
## person_brkloop_table 1380 11
## person_brkloop_table pId UniqueID OrigID FamName PerPre ID Father Mother Sex pedigree_link
person_link
##
## locus_table 1001 5
## locus_table pId LocusName Type AlleleCnt locus_link
##
## allele_table 2002 5
## allele_table pId AlleleName Frequency indexX locus_link
##
## marker_table 1000 7
## marker_table pId MarkerName pos_avg pos_female pos_male chromosome locus_link
##
## markerscheme_table 1000 4
## markerscheme_table pId key allele1 allele2
##
## map_table 2000 6
## map_table pId marker map position pos_female pos_male
##
## mapnames_table 2 6
## mapnames_table pId map sex_averaged_map male_sex_map female_sex_map name
##
## traitaff_table 1 4
## traitaff_table pId ClassCnt PenCnt locus_link

```

```
##
## affectclass_table 1 9
## affectclass_table pId MaleDef FemaleDef AutoDef MalePen FemalePen AutoPen locus_link
class_link
##
## phenotype_table 1380 4
## phenotype_table pId person_link bytes data
##
## genotype_table 1380 5
## genotype_table pId person_link chr bytes data
```

For each table generated, if 'verbose' is TRUE, we emit two lines: one with the number of rows and number of columns of the data frame and the other with the column names of the data frame.

Verbose Flag When verbose is set in the initial read.Mega2DB, the value will be remembered. It may be used by any subsequent function. If verbose is TRUE, Mega2R functions can print diagnostic information.

Use of an Environment The environment returned is used to store the data frames that contain the 'SQLite' database. The code above, stores the environment in global variable 'ENV'. If the named argument, 'envir', is not provided in any subsequent Mega2R function call, R will look up the value of 'ENV' in the global environment.

Back to more examples. The 'ls' function will show you all the variables in an environment. (You probably have used it without arguments to show you the variables in the .GlobalEnv.) Type:

```
ls(ENV)

## [1] "LocusCnt"          "MARKER_SCHEME"
## [3] "PhenoCnt"          "affectclass_table"
## [5] "allele_table"      "entrezGene"
## [7] "fam"               "int_table"
## [9] "locus_allele_table" "locus_table"
## [11] "map_table"         "mapnames_table"
## [13] "marker_table"      "markers"
## [15] "markerscheme_table" "pedigree_brkloop_table"
## [17] "pedigree_table"    "person_brkloop_table"
## [19] "person_table"      "phenotype_table"
## [21] "refIndices"        "refRanges"
## [23] "traitaff_table"    "txdb"
## [25] "unified_genotype_table" "verbose"
```

A more informative overview of the database can be had with:

```
showMega2ENV()
```

```
## locus count: 1001; phenotype count: 1; compression: 2 bits
```

```
## marker count: 1000; sample count: 1380
```

```
##
```

```
## genetic and physical maps:
```

```
## map name map number
```

```
## 1 Map 0
```

```
## 2 BP 1
```

```
##
```

```
## Phenotypes:
```

```
## Index Name Type
```

```
## 1 1 default affection
```

```
##
```

```
##
```

```
## basic tables:
```

```
## rows cols
```

```
## affectclass_table 1 9
```

```
## allele_table 2002 5
```

```
## int_table 34 3
```

```
## locus_table 1001 5
```

```
## map_table 2000 6
```

```
## mapnames_table 2 6
```

```
## marker_table 1000 8
```

```
## markerscheme_table 1000 4
```

```
## pedigree_brkloop_table 20 8
```

```
## pedigree_table 20 8
```

```
## person_brkloop_table 1380 11
```

```
## person_table 1380 11
```

```
## phenotype_table 1380 4
```

```
## traitaff_table 1 4
```

```
##
```

```
## derived tables:
```

```
## rows cols
```

```
## fam 1380 8
```

```
## markers 1000 5
```

```
## unified_genotype_table 1380 2
```

Use standard R operations to examine the created data frames Try typing:

```
str(ENV$locus_table)

## 'data.frame': 1001 obs. of 5 variables:
## $ pId      : int  1 2 3 4 5 6 7 8 9 10 ...
## $ LocusName : chr   "default" "snp1" "snp2" "snp3" ...
## $ Type      : int  2 4 4 4 4 4 4 4 4 4 ...
## $ AlleleCnt : int  2 2 2 2 2 2 2 2 2 2 ...
## $ locus_link: int  0 1 2 3 4 5 6 7 8 9 ...
```

Iterating through Gene Transcripts

There are two ways to compute a function on all the genotypes (or markers) in a transcript. These examples are explained more thoroughly in Section below.)

setRanges Mega2R has an internal default list of the chromosome and base pair ranges for many gene transcripts. These transcripts come from the UCSC Genome Browser reference assembly GRCH37. The list was further modified to eliminate multiple records from the same gene with the exact same transcript start and transcript end. The list contains about 29,000 records. You may load your own range set instead of the default. You create a data frame that contains at least a name "observation", a chromosome "observation", a start position "observation" and an end position "observation". And you create an integer vector that contains the column numbers of the chromosome "observation", the start position "observation" and the end position "observation". These two become the arguments to the *setRanges* function, viz.

```
setRanges(Ranges, Columns)
```

applyFnToRanges The function:

```
applyFnToRanges(DOcallback, envir = ENV)
```

goes through each transcript/range entry and finds the markers that fall within the bounds. It then builds a genotype matrix for the samples and each marker in the range.

The first argument of the *applyFnToRanges* function is a call back function. The function is called with four arguments: the genotype matrix, the markers in range, the selected transcript/range entry and the environment. The call back is invoked repeatedly for each transcript range that contains any markers.

If it is necessary to store information between successive invocations the environment ('envir') can be used.

applyFnToRanges can also be provided explicit ranges.

```
applyFnToRanges(DOcallback, ENV$refRanges, ENV$refIndices, envir = ENV)
applyFnToRanges(DOcallback, ENV$refRanges[ENV$refRanges$chromosome == 1,],
ENV$refIndices, envir = ENV)
```

setAnnotations Alternatively, you may search by genes. Here, the default transcript database is "TxDb.Hsapiens.UCSC.hg19.knownGene" from Bioconductor. Of course you can change it, by typing:

```
setAnnotations("txdb", "genedb")
```

where "txdb" is a string that is the name of transcript database that was fetched from Bioconductor, and similarly "genedb" is the name of a Bioconductor database that maps a gene id from the transcript to an entrez gene id. The *setAnnotations* command must be issued whenever R is started and the Mega2R library has been loaded. By way of a reminder, Section explains how to choose and install "TxDb.Hsapiens.UCSC.hg19.knownGene" or a different Annotation database.

Note: We could require that you add to your session

```
library(<name of the transcription database>)
```

but we have chosen not to. Instead, you just indicate the database name and the logic of *applyFnToGenes* does a 'loadNamespace' of the name string and then a 'get' of the name string from the namespace.

applyFnToGenes The function:

```
applyFnToGenes(DOcallback, genes_arg = c('DISP1', 'KIF26B', 'AK5', 'ST7L'),
envir = ENV)
```

will find all the transcripts for each gene requested from the Bioconductor database. It will then call *applyFnToRanges*: find all the markers in each transcript range, compute the genotype matrix and then invoke the call back function with the required four arguments.

The *applyFnToGenes* function has several other optional arguments that can request complete chromosomes, (multiple) ranges of base pairs on chromosomes, or collections of markers in addition to the *genes_arg* argument. All these arguments define ranges that are passed to *applyFnToRanges* for evaluation. Note: If the *genes_arg* argument is set to the special "gene" string "*", then all transcripts in the Bioconductor database, will be processed.

Iterating through the Genotypes

Encoded Genotypes The heart of the above two functions is the calculation of the genotype matrix of samples by markers containing the nucleotides (strings). The genotype information is most often stored in a compressed 2 bit representation. An Rcpp function does the conversion of the compressed genotype data to nucleotides. The function

```
getgenotypes(markers, envir = ENV)
```

returns the matrix for the specified markers. It can take one additional argument that supplies a string to separate the two alleles of each marker.

Raw Genotypes The function

```
getgenotypesraw(markers, envir = ENV)
```

is similar to the *getgenotypes* function except that the matrix it returns contains an integer encoding for each genotype. The integer's high 16 bits are the index for allele1 and the low 16 bits are the index for allele2.

Note: There are actually two different Rcpp functions for each named function in Section One function processes compressed genotype data and the other function processes uncompressed genotype data.

Using Mega2R to carry out automated gene-based association tests using 'pedgene'

Mega2R provides functions that permit one to run the 'pedgene' package to carry out gene-based association tests on family data looping over selected marker subsets.

Loading a Mega2 database with *init_pedgene*

Rather than read the Mega2 SQLite database with the *read.Mega2DB* function, described previously, here we use a specialized *init_pedgene* function to read the Mega2 database. This latter function calls a utility function also used by *read.Mega2DB*. Then it creates, edits, and rewrites the family data, storing it in a 'pedgene'-compatible data frame, **fam**. (**fam** merges data from the **pedigree_table**, **person_table** and **phenotype_table**.) *init_pedgene* purges persons with unknown case/control status which is necessary for the pedgene calculation. (When **fam** is updated, similar filtering is done to the persons in the **phenotype_table** and the **genotype_table**.) Finally, *init_pedgene* calculates some values that will be used repeatedly and stores them in the environment that is returned.

```
ENV = init_pedgene("seqsimr.db")
```

Gene Ranges reminder

Mega2R has an internal default list of the chromosome and base pair ranges for a number of gene transcripts. These transcripts come from the UCSC Genome Browser reference assembly GRCH37. The list was further modified to eliminate multiple records of the same gene with the exact same transcript start and transcript end. These data contain about 29,000 records. You may load your own range set instead of the default. You create a data frame that contains at least a name "observation", a chromosome "observation", a start position "observation" and an end position "observation". Then, you create an integer vector that contains the column numbers of the chromosome "observation", the start position "observation" and the end position "observation". These two objects are then arguments to the *setRanges* function, viz.

```
setRanges(Transcripts, Columns)
```

Gene transcripts reminder

If you plan to select transcripts by gene name, you must load them from Bioconductor. In Section , we indicated that you needed to type:

```
source("https://bioconductor.org/biocLite.R")
biocLite("TxDb.Hsapiens.UCSC.hg19.knownGene")
biocLite("org.Hs.eg.db")
```

just once to install the package. And then, to use the desired transcription data base, use this command from the Mega2R package as part of your session:

```
setAnnotations(txdb, entrezGene)
```

where 'txdb' is the name of Bioconductor transcription database, and 'entrezGene' is the name of Bioconductor mapping of gene name or gene alias to entrez gene id. Note: You do not need this command if you are using the default values shown above: TxDb.Hsapiens.UCSC.hg19.knownGene and org.Hs.eg.db.

Running pedgene on transcripts

By default, the function *Mega2pedgene* examines the first 100 transcripts and prints the results. (Internally, it calls the function *applyFnToRanges*.) For the seqsimr.db database, the first 100 transcripts finds only one transcript with several markers. (It is found because it is on chromosome 1 and it has a range that overlaps with some of the markers in our study.) To make this tutorial exercise run faster, we noticed that the identified transcript appeared at transcript 54; so we will restrict *Mega2pedgene* to a small range of transcripts around 54, viz. 50 through 60.

Note: 'verbose' needs to be TRUE, for the diagnostics to be printed.

```
ENV$verbose=TRUE
Mega2pedgene(gs=50:60)

## tryFn() No markers in range: NR_052010, IL11RA, chr9, 34653893, 34661898
## tryFn() No markers in range: NR_045785, CHIT1, chr1, 203185206, 203198860
## tryFn() No markers in range: NM_017799, TMEM260, chr14, 57046510, 57116232
## tryFn() No markers in range: NM_014705, DOCK4, chr7, 111366163, 111846462
## tryFn() No markers in range: NM_003759, SLC4A4, chr4, 72204769, 72437804
## tryFn() No markers in range: NM_002192, INHBA, chr7, 41728600, 41742706
## tryFn() No markers in range: NM_002202, ISL1, chr5, 50678957, 50690563
## tryFn() No markers in range: NM_003659, AGPS, chr2, 178257470, 178408564
```

```
## tryFn() No markers in range: NM_003658, BARX2, chr11, 129245880, 129322174

## tryFn() No markers in range: NM_018051, WDR60, chr7, 158649268, 158738883

## CEP104 snp24037 520 646 214
## CEP104 snp24039 940 386 54
## CEP104 snp24041 1377 3 0
## CEP104 snp24048 860 460 60
## CEP104 snp24494 789 501 90
## CEP104 snp24499 891 442 47
## CEP104 snp24506 891 442 47
## CEP104 snp24507 789 501 90

## chr gene nvariants start end sKernel_BT pKernel_BT sBurden_BT
## 1 chr1 CEP104 8 3728644 3773797 31.96998 0.6297541 -0.6496837

## pBurden_BT sKernel_MB pKernel_MB sBurden_MB pBurden_MB sKernel_UW
## 1 0.5158965 1184.995 0.5138866 -1.209563 0.2264468 222.8737

## pKernel_UW sBurden_UW pBurden_UW
## 1 0.4111136 -1.169488 0.242207
```

You will see many reports of "No markers in range", because the database only contains markers on a subrange of chromosome 1 whereas the transcripts span the entire genome. Occasionally you will see a listing of a gene name, markers, and count of 1/1, 1/2 and 2/2 genotypes, viz.

```
CEP104 snp24037 520 646 214
CEP104 snp24039 940 386 54
CEP104 snp24041 1377 3 0
CEP104 snp24048 860 460 60
CEP104 snp24494 789 501 90
CEP104 snp24499 891 442 47
CEP104 snp24506 891 442 47
CEP104 snp24507 789 501 90
```

The genotype matrix for these markers, along with the markers, the range used, and the environment are passed to the call back function *DOPedgene*. *DOPedgene* converts the genotype nucleotides 1/1, 1/2 and 2/2 to the values 0, 1 and 2 (or 2, 1, 0 if 1/1 is the genotype for the allele with the minor allele frequency). Then it runs 'pedgene'. The results are automatically stored in a data frame with "observations": prefix of chromosome, gene, number of markers and base pair range followed by 'pedgene' data: kernel and burden, value and p-values, four values for each of three weightings of the markers. These data are saved in the data frame, 'pedgene_results', in the environment. They are also printed when 'verbose' is TRUE.

Note: The results are always appended to the 'pedgene_results' data frame. You should truncate it when necessary.

You could run *Mega2pedgene* on all the transcript entries, but it takes a rather long time. You would type:

```
# we will skip this line for the document production because it takes too long
applyFnToRanges(DOpedgene, ENV$refRanges, ENV$refIndices, envir = ENV)
```

If you run the above test, you will see that genes *DISP1* and *KIF26B* have at least one p-value less than 0.01 and *AK5* and *STL7* at least one less than 0.03.

Running pedgene on selected genes

You may try searching for transcripts of specific genes. Here, the default transcript database is "TxDb.Hsapiens.UCSC.hg19.knownGene" from Bioconductor. Of course you can change it. Type:

```
setAnnotations("txdb", "genedb")
```

where "txdb" is a string that is the name of transcript database that was fetched from Bioconductor, and similarly "genedb" is the name of a Bioconductor database that maps a gene id from the input to an entrez gene id. You need to install any new database with biocLite, as shown earlier.

We leave the command below as an exercise. It runs a bit slowly. It needs to find all the transcripts for each gene, to find all the markers between each pair of transcript start/end ranges, to compute the genotype matrix for these markers and finally to call the *Vcallback* function with the appropriate arguments.

```
applyFnToGenes(DOpedgene, genes_arg = c('DISP1', 'KIF26B', 'AK5', 'STL7'), envir = ENV)
```

But let us run this function for a few genes:

```
applyFnToGenes(DOpedgene, genes_arg = c('DISP1', 'AK5'), envir = ENV)

##

## 'select()' returned 1:1 mapping between keys and columns
## 'select()' returned 1:many mapping between keys and columns
## tryFn() No markers in range: 26289, AK5, AK5, 1624, uc001dhm.2, 1, +,
77747662, 77780745

## AK5 snp80127 962 375 43
## AK5 snp80131 1095 272 13
## AK5 snp80142 1303 77 0
## AK5 snp80148 1345 35 0
## chr gene nvariants start end sKernel_BT pKernel_BT sBurden_BT
## 2 1 AK5 4 77747662 78025654 2593.945 0.228311 -1.400928
## pBurden_BT sKernel_MB pKernel_MB sBurden_MB pBurden_MB sKernel_UW
```

```
## 2 0.1612355 1899.249 0.06206979 -2.263002 0.02363554 175.4195
## pKernel_UW sBurden_UW pBurden_UW
## 2 0.051013 -2.290307 0.0220035
## DISP1 snp332780 1375 5 0
## DISP1 snp332781 675 559 146
## DISP1 snp332783 1365 15 0
## DISP1 snp332784 898 427 55
## chr gene nvariants start end sKernel_BT pKernel_BT
## 3 1 DISP1 4 222988431 223179337 4559.34 0.004224181
## sBurden_BT pBurden_BT sKernel_MB pKernel_MB sBurden_MB pBurden_MB
## 3 1.44652 0.1480315 5410.868 0.000253584 2.916462 0.003540261
## sKernel_UW pKernel_UW sBurden_UW pBurden_UW
## 3 277.508 0.04407319 2.094983 0.03617254
```

Note: The default behavior of *DOpedgene* is to append any new results to the end of the 'pedgene_results' data frame in the environment 'envir'.

Using Mega2R to carry out automated gene-based association tests using SKAT

Mega2R provides functions that permit one to run the SKAT package to carry out gene-based association tests on family data looping over selected marker subsets.

Loading a Mega2 database with *init.SKAT*

The 'init_SKAT' function is used to read the Mega2 database and initialize processing. This function calls the utility function, 'dbmega2_import', to read the database. Then it creates, edits, and rewrites the family data, storing it in the data frame, 'fam'. ('fam' contains data merged from the 'pedigree_table', 'person_table' and 'phenotype_table'.) 'init_SKAT' purges persons from the 'fam' data frame with unknown case/control status. 'setfam' sets 'fam' in the environment and insures that filtering is done to the phenotype_table and the genotype_table so they all have the same person_link key. Next, 'init_SKAT' decodes the phenotype_table into a simple data frame, 'phe'. You will definitely want to examine 'ENV'\$'phe' to choose which phenotype to use for the case/control. In addition, 'init_SKAT' calculates some values that will be used repeatedly and stores them in the environment that is returned. Finally, 'init_SKAT' stores the argument 'allMarkers' which tells later processing to ignore markers that show no variation (if FALSE). You may specify a "filename" argument to 'init_pedgeneSKAT' to indicate where to store the data frame that is produced later. (The default filename is NULL which indicates that the data frame should not be written.)

```
ENV = init_SKAT("seqsimr.db", filename = "SKAT.txt", verbose = F, allMarkers = F)
```

You can run the command with verbose equals TRUE to see details of the database as it is loaded:

```
ENV = init_SKAT("seqsimr.db", filename = "SKAT.txt", verbose = T, allMarkers = F)
```

Gene Ranges and Gene transcripts reminder

These data are contained in the section above.

Running Mega2SKAT on ranges

You should be familiar with the SKAT functions of the SKAT package before you read this section. The function, 'Mega2SKAT', is Mega2R's interface to SKAT, both to 'SKAT_Null_Model' and 'SKAT'. Its signature is:

```
Mega2SKAT = function (f, ty, gs = 1:100, skat = SKAT::SKAT, envir = ENV, ...)
```

The 'gs' argument indicates how many default range elements should be processed and the 'envir' argument specifies the '>environment' that contains all the Mega2R data frames.

Most of the time, before you call 'SKAT', you need to call 'SKAT_Null_Model' with a formula and an indicator for the type of the phenotype. 'Mega2SKAT' will take its first two arguments, a formula and a type (string) and call 'SKAT_Null_Model' with this information, viz.

```
SKAT_Null_Model(f, out_type = ty)
```

and store the results (in obj in the environment).

If the formula, 'f', is NULL, Mega2SKAT will not call 'SKAT_Null_Model' and you must do the equivalent before calling 'Mega2SKAT'. Store the result object in ENV\$obj. There are several reasons a custom call to the build the model could be necessary. You might want to use 'SKAT_Null_Model' but provided additional arguments viz. data, Adjustment, n.Resampling, type.Resampling. Alternatively, you might need to use a different model viz. SKAT_NULL_emmaX, SKAT_Null_Model_ChrX.

The 'skat' argument specifies the name of the SKAT package function to use; this is usually SKAT::SKAT, but could be SKAT::SKATBinary, SKAT::SKAT_CommonRare, etc. Any additional, arguments needed for the "skat" functions are provided to the 'Mega2SKAT' function and will be passed to the eventual call. All the "skat" functions are called with a genotype matrix for the markers, the object representing the Null Model and the additional arguments.

The Mega2R loop engine, iterates through each range and determines the set of markers contained. If there are no markers in a range, the genotype matrix is empty and Mega2SKAT will issue a warning. If a marker has no variation, Mega2SKAT will omit it, if the variable allMarkers is FALSE. Mega2SKAT will include it, if allMarkers is TRUE, but SKAT will more than likely issue a warning. The Mega2SKAT function also defines a callback function that converts the genotype information from the forms 1/1, 1/2, and 2/2 to 0, 1, and 2 with the major allele (flipped if necessary to be) 0 and then calls the specified "skat" function with the required arguments. Finally, the callback function stores the results.

By default, the function 'Mega2SKAT' examines the first 100 transcripts and prints the results. Note:

'verbose' is FALSE to eliminate the (excessive) diagnostics. A typical invocation of 'Mega2SKAT' could be:

```
ENV$verbose = FALSE
ENV$SKAT_results = ENV$SKAT_results[0, ]
Mega2SKAT(ENV$phe[, 3] - 1 ~ 1, "D", kernel = "linear.weighted",
          weights.beta=c(0.5,0.5))

## Sample size (non-missing y and X) = 1380, which is < 2000. The small sample adjustment is applied
```

These data are saved in the data frame, 'SKAT_results', in the environment. Note: The results are always appended to the data frame. You should truncate it when necessary.

```
print(ENV$SKAT_results)

##   chr  gene nvariants  start    end    skat
## 1 chr1 CEP104        8 3728644 3773797 0.495299
```

```
# we will skip this line for the Vignette document production because it takes too long
ENV$verbose = FALSE
ENV$SKAT_results = ENV$SKAT_results[0, ]
Mega2SKAT(ENV$phe[, 3] - 1 ~ 1, "D", kernel = "linear.weighted",
          weights.beta=c(0.5,0.5), gs=1:nrow(ENV$refRanges))
print(ENV$SKAT_results)
```

Running Mega2SKAT on genes

Let us run the Mega2SKAT function for a few genes:

```
ENV$SKAT_results = ENV$SKAT_results[0, ]
Mega2SKAT(ENV$phe[, 3] - 1 ~ 1, "D", kernel = "linear.weighted",
          weights.beta=c(0.5,0.5), genes = c('DISP1', 'AK5', 'KIF26B', 'ST7L'),
          envir = ENV)

## Sample size (non-missing y and X) = 1380, which is < 2000. The small sample adjustment is applied

## 'select()' returned 1:1 mapping between keys and columns
## 'select()' returned 1:many mapping between keys and columns
```

```
print(ENV$SKAT_results)
```

```
##      chr      gene nvariants      start      end      skat
## 1      1      AK5           4  77747662  78025654 0.0618782025
## 2      1     ST7L           2 113066141 113153625 0.0260893920
## 3      1  KIF26B           8 245318287 245809683 0.0100594906
## 4      1   DISP1           4 222988431 223179337 0.0008052555
```

Note: we set verbose to FALSE otherwise there will be a large number of print outs indicating no markers are in the range. (Recall the samples come from part of chromosome one, whereas the ranges cover the whole genome.) If you are adventurous you can look at all the transcript

```
ENV$verbose = FALSE
```

```
ENV$SKAT_results = ENV$SKAT_results[0, ]
```

```
Mega2SKAT(ENV$phe[, 3] - 1 ~ 1, "D", kernel = "linear.weighted",
           weights.beta=c(0.5,0.5), genes = '*', envir = ENV)
```

```
## Sample size (non-missing y and X) = 1380, which is < 2000. The small sample adjustment is applied
```

```
## 'select()' returned 1:many mapping between keys and columns
```

```
## 'select()' returned 1:1 mapping between keys and columns
```

Note: The default behavior of 'Mega2SKAT' is to append any new results to the end of the 'ENV\$SKAT_results' data frame.

```
print(ENV$SKAT_results)
```

```
##      chr      gene nvariants      start      end      skat
## 1      1   DDX11L1           8   11874   14409 0.084874638
## 2      1   NBPF20           4 144151519 144830407 0.157137831
## 3      1  GNG12-AS1           4  68297971  68668670 1.000000000
## 4      1  MIR548F1           4 186029867 186446655 0.732764650
## 5      1    TESK2           4  45809555  45956840 0.474770826
## 6      1   SDCCAG8           8 243449574 243663393 0.743744835
## 7      1    WDR63           8  85527993  85598821 1.000000000
## 8      1     CR1          12 207669473 207743990 0.774589848
## 9      1    GLIS1           4  53971906  54199877 0.453506643
## 10     1     DAB1           4  57463579  58716211 0.426490213
## 11     1     DPT           4 168664695 168698442 0.160401561
## 12     1     AK4           4  65613850  65697828 0.269995516
## 13     1    EPRS           8 220141942 220220000 1.000000000
```

## 14	1	NTNG1	4	107682629	108024475	0.273212746
## 15	1	ATF6	20	161736034	161933860	0.924534928
## 16	1	SRGAP2	8	206557369	206628046	0.092343492
## 17	1	AK5	4	77748287	78025654	0.062040512
## 18	1	BRINP3	4	190066797	190444942	1.000000000
## 19	1	LINC01140	4	87458690	87634886	1.000000000
## 20	1	KCNT2	4	196249674	196577499	0.513003364
## 21	1	TRABD2B	8	48226200	48462562	0.201653994
## 22	1	C1orf53	8	197871682	197876497	1.000000000
## 23	1	NCF2	4	183524697	183559739	0.391116249
## 24	1	NPR1	4	153651164	153666468	1.000000000
## 25	1	PLEKH01	4	150122170	150131825	0.656016802
## 26	1	PI4KB	8	151264273	151300191	0.423136031
## 27	1	GDAP2	4	118406107	118472302	0.305113749
## 28	1	ST7L	2	113084413	113160839	0.025964277
## 29	1	MARC2	16	220921676	220957596	0.358002461
## 30	1	KIF26B	8	245318287	245809683	0.010073914
## 31	1	HHAT	4	210501596	210849638	0.256823272
## 32	1	GATAD2B	8	153777203	153895451	0.872450869
## 33	1	LRRC7	4	70225858	70589171	0.543822765
## 34	1	LRRC7	4	70032868	70340687	0.908265470
## 35	1	RGS7	4	240938817	241519126	0.495229718
## 36	1	RYR2	8	237205702	237997288	0.497411200
## 37	1	SLC2A5	4	9097005	9129887	1.000000000
## 38	1	WASH7P	12	14362	16765	0.707964391
## 39	1	WASH7P	8	15603	29370	1.000000000
## 40	1	EVI5	4	93029199	93159945	0.328061334
## 41	1	TTC13	4	231044160	231114618	0.767719184
## 42	1	C1orf21	4	184356150	184598155	0.268918887
## 43	1	IGSF21	4	18434240	18704977	0.550668488
## 44	1	DISP1	4	222988431	223179337	0.000781078
## 45	1	CD2	4	117297086	117311851	0.097762438
## 46	1	C1orf158	8	12806163	12821102	0.495678067
## 47	1	NOS1AP	4	162039581	162326974	0.307191436
## 48	1	CEP104	8	3750235	3773797	0.492543861
## 49	1	CDK11B	4	1571100	1647917	0.783554451

Outputting Mega2R data to VCF format

The *Mega2VCF* function can output subsets of the Mega2R database as a VCF file, accompanied by related files (for phenotype and family data).

Writing a VCF File using Mega2R

First, create the directory "vcfr", so we can keep all the generated data in one place.

```
if (! dir.exists("vcfr")) dir.create("vcfr")
```

The *Mega2VCF* function takes an argument which is the prefix used on all generated files. We will make it a path that includes the directory **vcfr**. Below, we assume that a Mega2 database was stored in the environment, 'ENV'. If this is not the case, supply to *Mega2VCF* a named argument, 'envir', set to the environment you wish to use. (You will also have to change the references to the **markers** data frame, below, from 'ENV'\$markers to '<environment name>\$markers'.)

Note that Mega2VCF will create the .vcf file as well as a .fam file (in linkage format), a .phe file (phenotypes) and a few others. These files are listed below.

Creating the VCF file and related files is accomplished by typing:

```
Mega2VCF("vcfr/vcf.01", ENV$markers[ENV$markers$chromosome==1 ,])
```

Note: The line above places the data for only chromosome 1 in the files (Recall that our simulated data only contains chromosome 1). It is worth noting that the internal logic does not create a genotype matrix for all the requested markers at once, but rather works on a block of 1000 markers at a time, limiting the amount of memory required.

Note: In general, you can filter 'ENV\$markers' as needed. If the second argument is not present, all the markers are written out.

The generated files are listed below:

```
list.files("vcfr")

## [1] "vcf.01.fam" "vcf.01.freq" "vcf.01.map" "vcf.01.pen" "vcf.01.phe"
## [6] "vcf.01.vcf"
```

Converting Mega2R data to GenABEL format

Mega2R provides support for converting the data frames into 'GenABEL'

(<https://cran.r-project.org/package=GenABEL>) format (e.g. as gwaa.data-class objects).

Creating a GenABEL gwaa.data-class object using Mega2R

There are several data formats that Mega2 can read and transform into a database that are not directly accepted by GenABEL. But any Mega2 database that can be read into R, can be transformed to GenABEL following this example using **seqsimr.db**.

GenABEL (<http://www.genabel.org/>) can process PLINK .tped/.tfam files with the function 'convert.snp.tped' to create a GenABEL "raw" file. Then the "raw" file and a generated phenotype file can be processed by the function, 'load.gwaa.data', to yield a gwaa.data-class object. Currently, the function, *Mega2GenABEL*, uses this mechanism to convert Mega2 data frames to a gwaa.data-class object. *Mega2GenABEL* generates the names of the .tped/.fam/.phe files in the "tempdir()" directory with a prefix of "Mega2GenABEL". The GenABEL functions 'convert.snp.tped' and 'load.gwaa.data' are explained in <https://cran.r-project.org/package=GenABEL>. When *Mega2GenABEL* is finished, it deletes all the files that it generated.

If you have not done so already, install the 'GenABEL' package from CRAN.

Then type:

```
ENV = read.Mega2DB("seqsimr.db")
# This line converts the database to a gwaa.data-class object. The intermediate
# files in the tempdir() directory begin with "Mega2GenABEL"
seqsimgwaa = Mega2GenABEL()

## Reading individual ids from file '/var/folders/kn/h793jr1s0n39bdym6d7531qr0000gp/T//Rtmp0vTRGv/Mega2GenABELtped.raw'
## ... done. Read 1380 individual ids from file '/var/folders/kn/h793jr1s0n39bdym6d7531qr0000gp/T//Rtmp0vTRGv/Mega2GenABELtped.raw'
## Reading genotypes from file '/var/folders/kn/h793jr1s0n39bdym6d7531qr0000gp/T//Rtmp0vTRGv/Mega2GenABELtped.raw'
## ...done. Read 1000 SNPs from file '/var/folders/kn/h793jr1s0n39bdym6d7531qr0000gp/T//Rtmp0vTRGv/Mega2GenABELtped.raw'
## Writing to file '/var/folders/kn/h793jr1s0n39bdym6d7531qr0000gp/T//Rtmp0vTRGv/Mega2GenABELtped.raw'
## ... done.
## ids loaded...
## marker names loaded...
## chromosome data loaded...
## map data loaded...
## allele coding data loaded...
## strand data loaded...
## genotype data loaded...
## snp.data object created...
## assignment of gwaa.data object FORCED; X-errors were not checked!
```

You can use any of the GenABEL provided functions on the results, 'seqsimgwaa'.

```

str(seqsingwaa)

## Formal class 'gwaa.data' [package "GenABEL"] with 2 slots
##   ..@ phdata:'data.frame': 1380 obs. of  3 variables:
##   .. ..$ id      : chr [1:1380] "1SAP039_H05-0107" "1SAP039_H05-0106" "1SAP039_SAP039F14" "1SAP039_SAP039F14" ...
##   .. ..$ sex     : int [1:1380] 0 1 1 1 1 0 0 0 0 0 ...
##   .. ..$ default: int [1:1380] 1 1 1 1 1 1 1 1 1 1 ...
##   ..@ gtdata:Formal class 'snp.data' [package "GenABEL"] with 11 slots
##   .. ..@ nbytes   : num 345
##   .. ..@ nids     : int 1380
##   .. ..@ nsnp     : int 1000
##   .. ..@ idnames  : chr [1:1380] "1SAP039_H05-0107" "1SAP039_H05-0106" "1SAP039_SAP039F14" "1SAP039_SAP039F14" ...
##   .. ..@ snpname  : chr [1:1000] "snp1" "snp2" "snp3" "snp4" ...
##   .. ..@ chromosome: Factor w/ 1 level "1": 1 1 1 1 1 1 1 1 1 1 ...
##   .. ..@ attr(*, "names")= chr [1:1000] "snp1" "snp2" "snp3" "snp4" ...
##   .. ..@ map      : Named num [1:1000] 2 3 4 5 201 ...
##   .. ..@ attr(*, "names")= chr [1:1000] "snp1" "snp2" "snp3" "snp4" ...
##   .. ..@ coding   :Formal class 'snp.coding' [package "GenABEL"] with 1 slot
##   .. ..@ .Data: raw [1:1000] 01 01 01 01 ...
##   .. ..@ strand   :Formal class 'snp.strand' [package "GenABEL"] with 1 slot
##   .. ..@ .Data: raw [1:1000] 00 00 00 00 ...
##   .. ..@ male     : Named int [1:1380] 0 1 1 1 1 0 0 0 0 0 ...
##   .. ..@ attr(*, "names")= chr [1:1380] "1SAP039_H05-0107" "1SAP039_H05-0106" "1SAP039_SAP039F14" "1SAP039_SAP039F14" ...
##   .. ..@ gtps     :Formal class 'snp.mx' [package "GenABEL"] with 1 slot
##   .. ..@ .Data: raw [1:345, 1:1000] 59 55 55 55 ...

```

Regression testing

To verify that the functions described above have worked correctly, we can compare their output files to those created by Mega2, itself, to verify that they are identical, as expected. This type of testing is known as regression testing.

VCF regression

If you have a executable copy of Mega2, you can run the code below using the MEGA2.BATCH.vcf file and provided example database.

Cut and paste the shell code and you should see similar results to what is presented here.

Otherwise, the text below is an illustration of what should happen. These lines use the Mega2 executable to populate the **vcf** directory with the same set of files that the were created in the **vcfr** directory.

At the Unix command prompt, type:

```
mkdir vcf
mega2 MEGA2.BATCH.vcf
```

Note: This document will not invoke *mega2*, but use the *vcf* directory.

Note: The vignette associated with the Mega2R CRAN package illustrates what this Mega2 run would look like.

The abbreviated MEGA2.BATCH.vcf file is below. (The initial comment section is not shown and unimportant lines are elided.) Notice that there are no INPUT FILES, just a database file.)

```
...
Output_Path=vcf
...
Analysis_Option=VCF
...
DBfile_name=seqsimr.db
Chromosome_Single=1
...
file_name_stem=vcf
...
```

You should compare the two directories: **vcf** and **vcfr**. Please add the `-w` flag to the “diff” command. This causes the comparison to ignore differences in white space. Lines that contain dates may be different between the two directories. At a Unix command prompt, type:

```
diff -wrs vcf vcfr
```

Note: To make this tutorial only dependent on R, the above code is not actually run. And the results, shown below, were captured from an environment where we had both R and the Mega2 executable available.

The *diff* should show:

```
## Files vcf/vcf.01.fam and vcfr/vcf.01.fam are identical
## Files vcf/vcf.01.freq and vcfr/vcf.01.freq are identical
## Files vcf/vcf.01.map and vcfr/vcf.01.map are identical
## Files vcf/vcf.01.pen and vcfr/vcf.01.pen are identical
## Files vcf/vcf.01.phe and vcfr/vcf.01.phe are identical
## Files vcf/vcf.01.vcf and vcfr/vcf.01.vcf are identical
```

GenABEL regression

This test is a bit more complicated. We intend to verify that the object generated by *Mega2GenABEL* is the same as the GenABEL object we started with. First, we load ‘GenABEL’ and access its data:

```
library(GenABEL)
data(srdta)
```

Then we dump the GenABEL data as PLINK .ped/.map/.phe files

```
dmpPed(gwaa_ = srdta, pfx = "srdta")
```

To produce the Mega2 database 'srdta.db', we switch to the Unix command line and run:

```
mega2 MEGA2.BATCH.srdta
```

Note: This document will not invoke *mega2*, but use the *srdta.db* database that is in this package.

Note: The vignette associated with the Mega2R CRAN package illustrates what this Mega2 run would look like.

The MEGA2.BATCH.srdta file The abbreviated MEGA2.BATCH.srdta file is below. (The initial comment section is not shown and unimportant lines are elided.)

```
Input_Database_Mode=1
Input_Format_Type=4
Input_Pedigree_File=srdta.ped
Input_PLINK_Map_File=srdta.map
Input_PhenoType_File=srdta.phe
...
Analysis_Option=Dump
...
DBfile_name=srdta.db
...
```

Note: We have provided a copy of the *srdta.db* database in the Mega2R package. You can proceed with the steps that follow, even if you did not create your own database.

Now, we go back into R and type:

```
ENV = read.Mega2DB("srdta.db")
mega = Mega2GenABEL()

## Reading individual ids from file '/var/folders/kn/h793jr1s0n39bdym6d7531qr0000gp/T//Rtmp0vTRGv/Mega2GenABELped.ra
## ... done. Read 2500 individual ids from file '/var/folders/kn/h793jr1s0n39bdym6d7531qr0000gp/T//Rtmp0vTRGv/Mega2GenABELped.ra
## Reading genotypes from file '/var/folders/kn/h793jr1s0n39bdym6d7531qr0000gp/T//Rtmp0vTRGv/Mega2GenABELped.ra
## ...done. Read 833 SNPs from file '/var/folders/kn/h793jr1s0n39bdym6d7531qr0000gp/T//Rtmp0vTRGv/Mega2GenABELped.ra
## Writing to file '/var/folders/kn/h793jr1s0n39bdym6d7531qr0000gp/T//Rtmp0vTRGv/Mega2GenABELped.ra'
```

```
## ... done.
## ids loaded...
## marker names loaded...
## chromosome data loaded...
## map data loaded...
## allele coding data loaded...
## strand data loaded...
## genotype data loaded...
## snp.data object created...
## assignment of gwaa.data object FORCED; X-errors were not checked!
```

'mega' should be the same as the 'srdta' gwaa.data-class object. You can compare them however you prefer. For example, in R, type:

```
str(mega)

## Formal class 'gwaa.data' [package "GenABEL"] with 2 slots
##   ..@ phdata:'data.frame': 2500 obs. of  8 variables:
##   .. ..$ id      : chr [1:2500] "p1_p1" "p2_p2" "p3_p3" "p4_p4" ...
##   .. ..$ sex     : int [1:2500] 1 1 0 1 1 0 0 1 0 0 ...
##   .. ..$ age     : num [1:2500] 43.4 48.2 37.9 53.8 47.5 45 52 42.5 29.7 45.8 ...
##   .. ..$ qt1     : num [1:2500] -0.58 0.8 -0.52 -1.55 0.25 0.15 -0.56 -9 -2.26 -1.32 ...
##   .. ..$ qt2     : num [1:2500] 4.46 6.32 3.26 888 5.7 4.65 4.64 5.77 0.71 3.26 ...
##   .. ..$ qt3     : num [1:2500] 1.43 3.9 5.05 3.76 2.89 1.87 2.49 2.68 1.45 0.85 ...
##   .. ..$ bt      : int [1:2500] 0 1 1 1 1 0 0 1 0 0 ...
##   .. ..$ default: int [1:2500] 0 1 1 1 1 0 0 1 0 0 ...
##   ..@ gtdata:Formal class 'snp.data' [package "GenABEL"] with 11 slots
##   .. .. ..@ nbytes      : num 625
##   .. .. ..@ nids        : int 2500
##   .. .. ..@ nsnp      : int 833
##   .. .. ..@ idnames     : chr [1:2500] "p1_p1" "p2_p2" "p3_p3" "p4_p4" ...
##   .. .. ..@ snpnames    : chr [1:833] "rs10" "rs18" "rs29" "rs65" ...
##   .. .. ..@ chromosome: Factor w/ 1 level "1": 1 1 1 1 1 1 1 1 1 1 ...
##   .. .. ..- attr(*, "names")= chr [1:833] "rs10" "rs18" "rs29" "rs65" ...
##   .. .. ..@ map        : Named num [1:833] 2500 3500 5750 13500 14250 ...
##   .. .. ..- attr(*, "names")= chr [1:833] "rs10" "rs18" "rs29" "rs65" ...
##   .. .. ..@ coding     :Formal class 'snp.coding' [package "GenABEL"] with 1 slot
##   .. .. .. ..@ .Data: raw [1:833] 08 0b 0c 07 ...
##   .. .. ..@ strand     :Formal class 'snp.strand' [package "GenABEL"] with 1 slot
##   .. .. .. ..@ .Data: raw [1:833] 00 00 00 00 ...
```

```
## .. ..@ male      : Named int [1:2500] 1 1 0 1 1 0 0 1 0 0 ...
## .. ..- attr(*, "names")= chr [1:2500] "p1_p1" "p2_p2" "p3_p3" "p4_p4" ...
## .. ..@ gtps      :Formal class 'snp.mx' [package "GenABEL"] with 1 slot
## .. ..- attr(*, "names")= chr [1:2500] "p1_p1" "p2_p2" "p3_p3" "p4_p4" ...

str(srdta)

## Formal class 'gwaa.data' [package "GenABEL"] with 2 slots
## ..@ phdata:'data.frame': 2500 obs. of  7 variables:
## ..$ id : chr [1:2500] "p1" "p2" "p3" "p4" ...
## ..$ sex: int [1:2500] 1 1 0 1 1 0 0 1 0 0 ...
## ..$ age: num [1:2500] 43.4 48.2 37.9 53.8 47.5 45 52 42.5 29.7 45.8 ...
## ..$ qt1: num [1:2500] -0.58 0.8 -0.52 -1.55 0.25 0.15 -0.56 NA -2.26 -1.32 ...
## ..$ qt2: num [1:2500] 4.46 6.32 3.26 888 5.7 4.65 4.64 5.77 0.71 3.26 ...
## ..$ qt3: num [1:2500] 1.43 3.9 5.05 3.76 2.89 1.87 2.49 2.68 1.45 0.85 ...
## ..$ bt : int [1:2500] 0 1 1 1 1 0 0 1 0 0 ...
## ..@ gtdata:Formal class 'snp.data' [package "GenABEL"] with 11 slots
## ..$ nbytes : num 625
## ..$ nids : int 2500
## ..$ nsnp : int 833
## ..$ idnames : chr [1:2500] "p1" "p2" "p3" "p4" ...
## ..$ snpname : chr [1:833] "rs10" "rs18" "rs29" "rs65" ...
## ..$ chromosome: Factor w/ 1 level "1": 1 1 1 1 1 1 1 1 1 1 ...
## ..$ attr(*, "names")= chr [1:833] "rs10" "rs18" "rs29" "rs65" ...
## ..$ map : Named num [1:833] 2500 3500 5750 13500 14250 ...
## ..$ attr(*, "names")= chr [1:833] "rs10" "rs18" "rs29" "rs65" ...
## ..$ coding :Formal class 'snp.coding' [package "GenABEL"] with 1 slot
## ..$ .Data: raw [1:833] 08 0b 0c 03 ...
## ..$ strand :Formal class 'snp.strand' [package "GenABEL"] with 1 slot
## ..$ .Data: raw [1:833] 01 01 02 01 ...
## ..$ male : Named int [1:2500] 1 1 0 1 1 0 0 1 0 0 ...
## ..$ attr(*, "names")= chr [1:2500] "p1" "p2" "p3" "p4" ...
## ..$ gtps :Formal class 'snp.mx' [package "GenABEL"] with 1 slot
## ..$ .Data: raw [1:625, 1:833] 55 59 55 a5 ...
```

You can compare these printouts by eye. Alternatively, the function *Mega2GenABELtst()* can be used to compare the phenotype data, genotype data, and object metadata, item by item.

```

options(max.print = 30)
Mega2GenABELtst(mega_ = mega, gwaa_ = srdta)

## all(mega_@phdata$sex == gwaa_@phdata$sex) [1] TRUE
## all(mega_@phdata$age == gwaa_@phdata$age) [1] TRUE
## all(mega_@phdata$qt1 == gwaa_@phdata$qt1) [1] TRUE
## all(mega_@phdata$qt2 == gwaa_@phdata$qt2) [1] TRUE
## all(mega_@phdata$qt3 == gwaa_@phdata$qt3) [1] TRUE
## all(mega_@phdata$bt == gwaa_@phdata$bt) [1] TRUE
## all(mega_@gtdata@nids == gwaa_@gtdata@nids)[1] TRUE
## all(mega_@gtdata@nsnps == gwaa_@gtdata@nsnps)[1] TRUE
## all(mega_@gtdata@nbytes == gwaa_@gtdata@nbytes)[1] TRUE
## all(mega_@gtdata@idnames == gwaa_@gtdata@idnames)[1] FALSE
## all(mega_@gtdata@snpsnames == gwaa_@gtdata@snpsnames)[1] TRUE
## all(mega_@gtdata@chromosome == gwaa_@gtdata@chromosome)[1] TRUE
## all(mega_@gtdata@map == gwaa_@gtdata@map)[1] TRUE
## all(mega_@gtdata@male == gwaa_@gtdata@male)[1] TRUE
## all(mega_@gtdata@coding == gwaa_@gtdata@coding)[1] FALSE
## all(mega_@gtdata@strand == gwaa_@gtdata@strand)[1] FALSE
## all(mega_@gtdata@gtps == gwaa_@gtdata@gtps)[1] FALSE
## all(mega_@gtdata == gwaa_@gtdata)[1] TRUE
## [1] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [1] FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE FALSE
## [1] "markers that differ"
## [1] "markers that differ"
##      locus_link locus_link_fill MarkerName chromosome position
## 4           9           9      rs65           1      13500
## 8           13           13     rs130           1      27250
## 10          15           15     rs150           1      33250
## 20          25           25     rs324           1      82500
## 23          28           28     rs348           1      88000
## 24          29           29     rs361           1      90000
## [ reached getOption("max.print") -- omitted 209 rows ]
## [1] "markerscheme_table (compression) alleles that differ"
## [1] "markerscheme_table (compression) alleles that differ"
##      pId key allele1 allele2
## 4      4  9      1      2
## 8      8 13      1      2
## 10     10 15      1      2

```

```
## 20 20 25 2 1
## 23 23 28 2 1
## 24 24 29 1 2
## 29 29 34 1 2
## [ reached getOption("max.print") -- omitted 208 rows ]
## [1] "allele values for markers that differ"
## [1] "allele values for markers that differ"
##      locus_link pId.x AlleleName.x Frequency.x indexX.x pId.y AlleleName.y
## 4           9 19           A 0.28027754      1 20           T
## 8          13 27           A 0.30622110      1 28           G
## 10         15 31           A 0.65998312      1 32           C
##      Frequency.y indexX.y
## 4 0.71972246      2
## 8 0.69377890      2
## 10 0.34001688      2
## [ reached getOption("max.print") -- omitted 212 rows ]
```

These results are somewhat disappointing. But ...

First, "mega2" prefers to use the pedigree id concatenated to the person id as the sample name, even if the person id itself is unique.

The second difference, in **gtdata@strand**, occurs because Mega2 does not keep track of strand orientation and always returns 0.

The final difference is more subtle. The **gtdata** comparison line decodes the genotype to characters then normalizes the two heterozygous cases to one value. These results are the same for the two different *gwaa*.class-objects; hence the data are really the same. But *Mega2GenABEL* converts the data frames using the 'convert.snp.tped' function while the 'srdata' object, generated by GenABEL, uses the 'convert.snp.text'. Though both conversion functions encode the genotype to a number, the 'convert.snp.tped' function rearranges the genotype so that the major allele appears first, the 'convert.snp.text' does no such rearrangement. Because **gtdata@coding** and **gtdata@gtps** compare the encoded allele pairs and corresponding encoded genotypes, they sometimes differ.

```
clean_mega2rtutorial_data(".", " ")
```

Summary

As with all software projects, **Mega2R** is a work in progress, and so there are a number of possible improvements that could be made in the future. These include:

- defer reading the **genotype_table** until the set of chromosomes that are needed can be computed and then only read the required chromosome genotype vectors.

Availability and requirements

Project name: **Mega2R**

Project home page: <https://watson.hgen.pitt.edu/register>

Project BitBucket repository: <https://bitbucket.org/dweeks/mega2/>

Project documentation page: <https://watson.hgen.pitt.edu/register/docs/mega2R.html>

Operating system(s): Platform independent

Programming language: R, C++

Other requirements; R, SQLite library

License: GNU GPL 2

List of abbreviations used

Competing interests

The authors declare that they have no competing interests.

Authors' contributions

RVB designed, debugged, and maintains Mega2R; DEW helped prepare the Mega2R R package for submission to CRAN. RVB and JRS maintain and extend Mega2. DEW provided inspiration and provided the problems that Mega2R solves. RVB, with assistance from DEW, wrote this manuscript. All authors contributed to the review of the manuscript and agreed to the final content.

Acknowledgments

This work was supported by NIH grant R01 GM076667 (PI: Weeks). Earlier contributions to our Mega2 code base were made by Charles P. Kollar, Nandita Mukhopadhyay, Lee Almasy, Mark Schroeder, and William P. Mulvihill.

References

- [1] Mukhopadhyay N, Almasy L, Schroeder M, Mulvihill W, Weeks D: **Mega2: data-handling for facilitating genetic linkage and association analyses**. *Bioinformatics* 2005, **21**(10):2556–7.
- [2] Baron RV, Kollar C, Mukhopadhyay N, Weeks DE: **Mega2: validated data-reformatting for linkage and association analyses**. *Source Code Biol Med* 2014, **9**:26.
- [3] Lathrop GM, Lalouel JM: **Easy calculations of lod scores and genetic risks on small computers**. *Am J Hum Genet* 1984, **36**(2):460–5.

- [4] Lathrop GM, Lalouel JM, White RL: **Construction of human linkage maps: likelihood calculations for multilocus linkage analysis.** *Genet Epidemiol* 1986, **3**:39–52.
- [5] Lathrop GM, Lalouel JM: **Efficient computations in multilocus linkage analysis.** *Am J Hum Genet* 1988, **42**(3):498–505.
- [6] Purcell S, Neale B, Todd-Brown K, Thomas L, Ferreira MAR, Bender D, Maller J, Sklar P, de Bakker PIW, Daly MJ, Sham PC: **PLINK: a tool set for whole-genome association and population-based linkage analyses.** *Am J Hum Genet* 2007, **81**(3):559–75.
- [7] Danecek P, Auton A, Abecasis G, Albers CA, Banks E, DePristo MA, Handsaker RE, Lunter G, Marth GT, Sherry ST, McVean G, Durbin R, 1000 Genomes Project Analysis Group: **The variant call format and VCFtools.** *Bioinformatics* 2011, **27**(15):2156–8.
- [8] Howie BN, Donnelly P, Marchini J: **A flexible and accurate genotype imputation method for the next generation of genome-wide association studies.** *PLoS Genet* 2009, **5**(6):e1000529.
- [9] Howie B, Marchini J, Stephens M: **Genotype imputation with thousands of genomes.** *G3 (Bethesda)* 2011, **1**(6):457–70.
- [10] Howie B, Fuchsberger C, Stephens M, Marchini J, Abecasis GR: **Fast and accurate genotype imputation in genome-wide association studies through pre-phasing.** *Nat Genet* 2012, **44**(8):955–9.
- [11] Marchini J, Howie B: **Genotype imputation for genome-wide association studies.** *Nat Rev Genet* 2010, **11**(7):499–511.
- [12] R Core Team: *R: A Language and Environment for Statistical Computing.* R Foundation for Statistical Computing, Vienna, Austria 2017, [<https://www.R-project.org/>].
- [13] Aulchenko Y, Ripke S, Isaacs A, van Duijn C: **GenABEL: an R package for genome-wide association analysis.** *Bioinformatics* 2007, **23**(10):1294–6.
- [14] Schaid D, McDonnell S, Sinnwell J, Thibodeau S: **Multiple Genetic Variant Association Testing by Collapsing and Kernel Methods With Pedigree or Population Structured Data.** *Genet Epidemiol* 2013, **37**(5):409–418.
- [15] Wu MC, Lee S, Cai T, Li Y, Boehnke M, Lin X: **Rare Variant Association Testing for Sequencing Data Using the Sequence Kernel Association Test (SKAT).** *American Journal of Human Genetics* 2011, **89**:82–93.
- [16] Lee S, Emond M, Bamshad M, Barnes K, Rieder M, Nickerson D, Team NGESPEL, Christiani D, Wurfel M, Lin X: **Optimal unified approach for rare variant association testing with application to small sample case-control whole-exome sequencing studies.** *American Journal of Human Genetics* 2012, **91**:224–237.
- [17] Lee S, Wu MC, Lin X: **Optimal tests for rare variant effects in sequencing association studies.** *Biostatistics* 2012, **13**:762–775.

- [18] **VCF Format (early spec)** [<http://www.internationalgenome.org/wiki/Analysis/Variant%20Call%20Format/vcf-variant-call-format-version-40/>].
- [19] **1000 Genomes** [<http://www.internationalgenome.org/home>].
- [20] **VCF Format** [<http://samtools.github.io/hts-specs/>].
- [21] Xie Y: *knitr: A General-Purpose Package for Dynamic Report Generation in R* 2017, [<https://yihui.name/knitr/>]. [R package version 1.17].
- [22] Xie Y: **knitr: A Comprehensive Tool for Reproducible Research in R**. In *Implementing Reproducible Computational Research*. Edited by Stodden V, Leisch F, Peng RD, Chapman and Hall/CRC 2014 [<http://www.crcpress.com/product/isbn/9781466561595>]. [ISBN 978-1466561595].
- [23] Xie Y: *Dynamic Documents with R and knitr*. Boca Raton, Florida: Chapman and Hall/CRC, 2nd edition 2015, [<https://yihui.name/knitr/>]. [ISBN 978-1498716963].
- [24] Chung R, Tsai W, Hsieh C, Hung K, Hsiung C, Hauser E: **SeqSIMLA2: simulating correlated quantitative traits accounting for shared environmental effects in user-specified pedigree structure**. *Genet Epidemiol*. 2015, **39**:20–4.

Illustrations and figures

Figure captions

Additional files

Appendix

The data from most of the Mega2 SQLite tables are potentially useful and are made available in data frames.

Miscellaneous Tables

int_table

This table contains data like Phenotype Count and Locus Count; they can be calculated from tables but its faster to have the value available. There also are a **double_table**, **charstar_table** and **stuff_table** for storing: double float, character string and byte vector values, respectively.

column	contents
pId	unique database table key
key	name of an integer
value	value for integer

Pedigree & Person Tables

pedigree_table (and pedigree_brkloop_table)

This table provides the basic data for a family: name, an index and count of members.

column	contents
pId	unique database table key
Num	unique ID for pedigree
EntryCnt	number of persons in pedigree
Name	Mega2 alternate name
PedPre	ID from study
OriginalID	Mega2 alternate name
origped	Mega2 alternate name
pedigree_link	unique integer linking family info

person_table (and person_brkloop_table)

This table provides the basic data for a person: name, an index, parents, and sex.

column	contents
pId	unique database table key
UniqueID	unique ID for person
OrigID	Mega2 alternate name
FamName	Mega2 alternate family name
PerPre	ID from study
ID	numeric ID for person
Father	father's ID
Mother	mother's ID
Sex	person's sex
pedigree_link	unique integer linking family info
person_link	unique integer linking person info

pedigree.brkloop.table and person.brkloop.table

These tables are available for the analyses that do not accept pedigrees with loops. The loops are broken and these tables create a “doppelganger” for one person in each loop and adjust the parents of one of the pair to break the loop. Compared to the pedigree_table and person_table, there will be an extra person in some families (those with loops).

Marker Information Tables

locus.table

This table provides the basic data for a locus: its name, type and allele count.

column	contents
pId	unique database table key
LocusName	locus (phenotype/genotype) name
Type	QUANT, AFFECTION, BINARY, NUMBERED, XLINKED, YLINKED
AlleleCnt	number of alleles for this marker
locus_link	unique integer linking loci info

allele.table

This table list the allele data for each locus: name, frequency and a index increasing monotonically within the locus.

column	contents
pId	unique database table key
AlleleName	allele nucleotide
Frequency	allele frequency
indexX	numeric index of allele 1-N
locus_link	unique integer linking loci info

marker_table

This table provides the basic data for a marker: its name, chromosome and position.

column	contents
pId	unique database table key
MarkerName	name of marker
pos_avg	preferred average genetic position
pos_female	preferred female genetic position
pos_male	preferred male genetic position
chromosome	chromosome of marker
locus_link	unique integer linking loci info

markerscheme_table

This table is used as part of the compression algorithm to compress nucleotide values.

column	contents
pId	unique database table key
key	unique integer linking loci
allele1	allele 1 value
allele2	allele 2 value

map_table

This table provides the genetic position values for a given marker according to the “map” index defined in the table below.

column	contents
pId	unique database table key
marker	unique integer linking loci
map	map index number
position	average genetic position
pos_female	female genetic position
pos_male	male genetic position

mapnames_table

This table provides a mapping of a map name string to a map index. If all of *sex_averaged_map*, *male_sex_map* and *female_sex_map* are 0, the corresponding map's *position* specifies a base pair position.

column	contents
pId	unique database table key
map	map index number
sex_averaged_map	1 if present else 0
male_sex_map	1 if present else 0
female_sex_map	1 if present else 0
name	map name

Phenotype Tables**traitaff_table**

This table indicates the number of classes and number of penetrance entries for each affection locus.

column	contents
pId	unique database table key
ClassCnt	number of liability classes
PenCnt	number of penetrance values
locus_link	unique integer linking loci info

affectclass_table

This table holds penetrance data for each loci for each class.

column	contents
pId	unique database table key
MaleDef	0 indicates corresponding penetrance was set
FemaleDef	0 indicates corresponding penetrance was set
AutoDef	0 indicates corresponding penetrance was set
MalePen	male penetrance
FemalePen	female penetrance
AutoPen	autosomal penetrance
locus_link	unique integer linking loci info
class_link	unique integer specifying class index

phenotype_table

This table holds phenotype data for each person.

column	contents
pId	unique database table key
person_link	unique integer linking person info
bytes	byte count of data
data	raw data

Genotype Table**genotype_table**

This table holds genotype data for each person.

column	contents
pId	unique database table key
person_link	unique integer linking person info
chr	chromosome number
bytes	byte count of data
data	raw data